



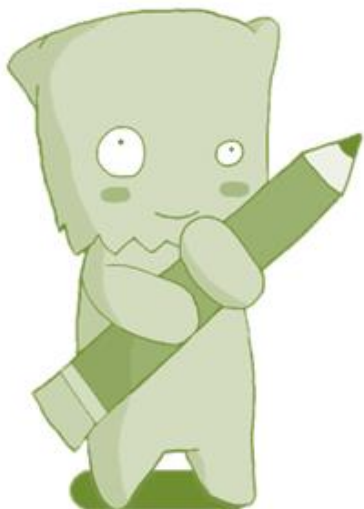
Chapter 1

Role of Statistics





- ♣ Engineering Method and Statistical Thinking
- ♣ Collecting Engineering Data
- ♣ Mechanistic and Empirical Models
- ♣ Observing Processes Over Time



What Do Engineers Do?

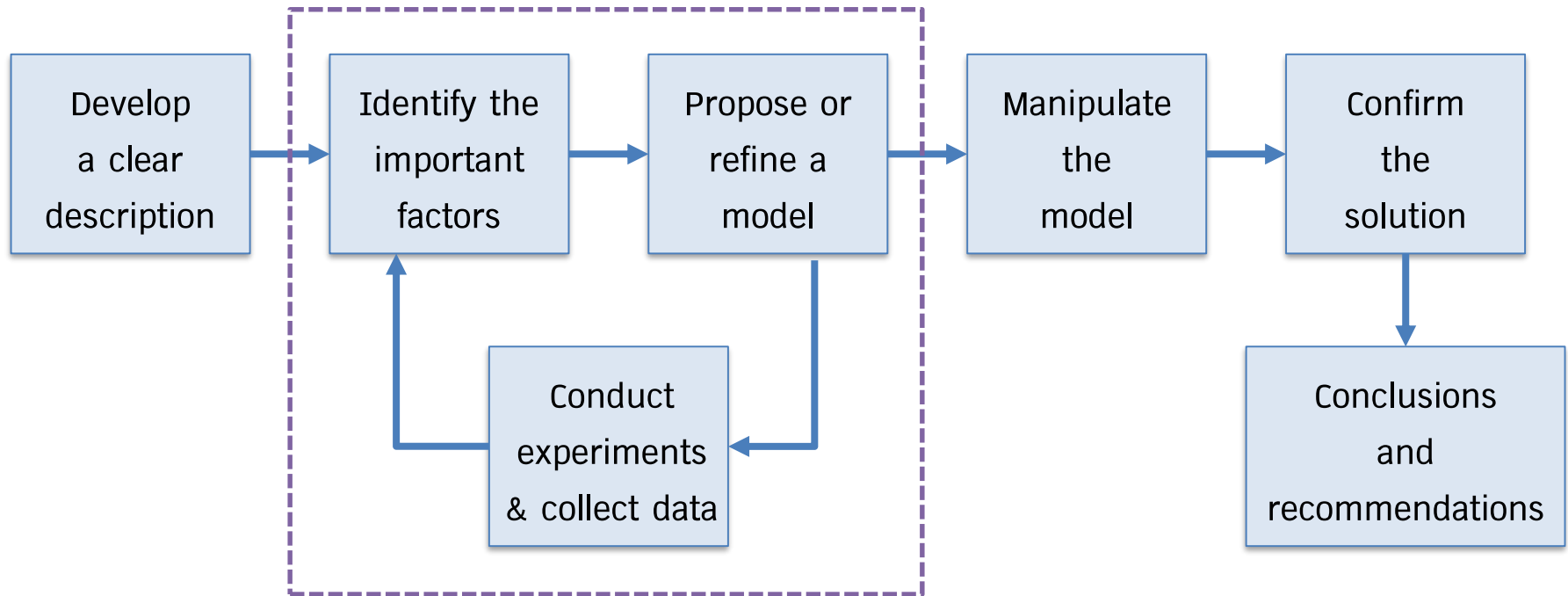
An **engineer** is someone who solves problems of interest to society with the efficient application of scientific principles by:

- ◆ Refining existing products
- ◆ Designing new products or processes





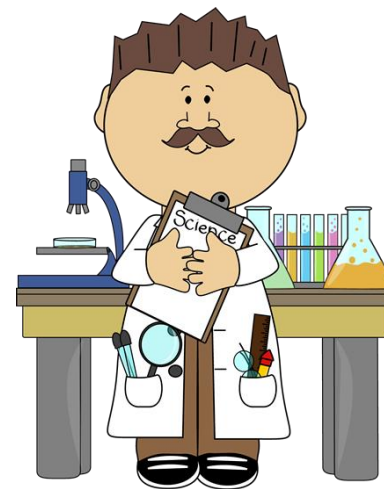
The Creative Process



The engineering method
for problem-solving

Statistics Support Creative Process

- ♣ Statistics is the **science of data**.
- ♣ The field of statistics deals with the **collection**, **presentation**, **analysis**, and **use of data** to:
 - ◆ make decisions
 - ◆ solve problems
 - ◆ design products and processes





Experiments & Processes Are Not Deterministic

- ♣ Successive observations of a system or phenomenon **do not produce exactly** the same result.
- ♣ Statistical techniques are useful for describing and understanding **variability**.
- ♣ Statistics gives us a framework for **describing this variability** and for learning about potential **sources of variability**.



Example of Variability

- ♣ An engineer is designing a **nylon connector** to be used in an automotive engine application.
- ♣ If the pull-off force is too low, the connector may fail when it is installed in an engine.
- ♣ The engineer is considering establishing the design specification on wall **thickness at 3/32 inch**, but is somewhat uncertain about the effect of this decision on the connector **pull-off force**.
- ♣ **Eight prototype units** are produced and their pull-off forces measured (in pounds):
12.6, 12.9, 13.4, 12.3, 13.6, 13.5, 12.6, 13.1





Example of Variability (Cont.)

- ♣ The **dot diagram** is a very useful plot for displaying a small body of data - *say up to about 20 observations*.
- ♣ This plot allows us to see easily two features of the data; the **location** or the middle, and the **scatter** or **variability**.

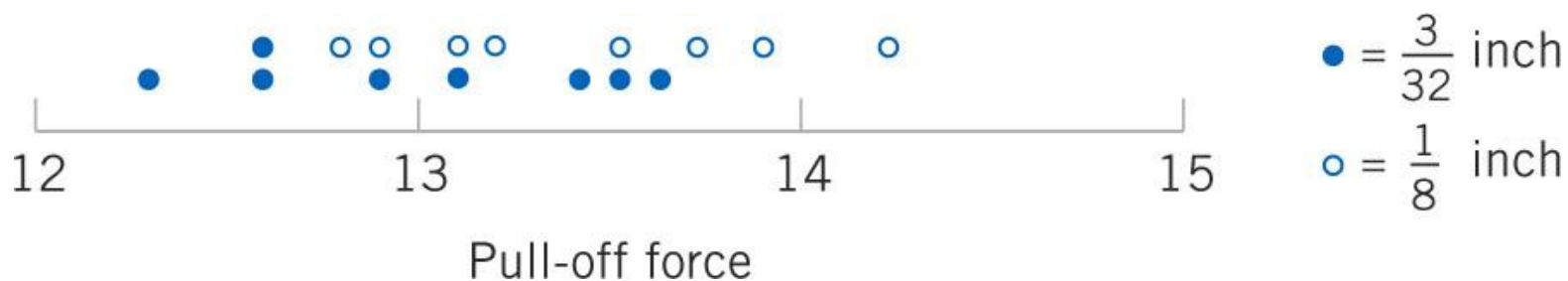


Dot diagram of the pull-off force data when wall thickness is 3/32 inch.



Example of Variability (Cont.)

- ♣ The engineer considers **an alternate design** and eight prototypes are built and pull-off force measured.
- ♣ The dot diagram can be used to compare two sets of data.



Dot diagram of pull-off force
for two wall thicknesses.



Example of Variability (Cont.)

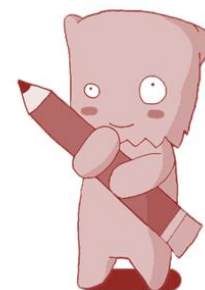
- ♣ Since pull-off force varies or exhibits variability, it is a **random variable**.
- ♣ A random variable, X , can be modeled by:

$$X = \mu + \varepsilon$$

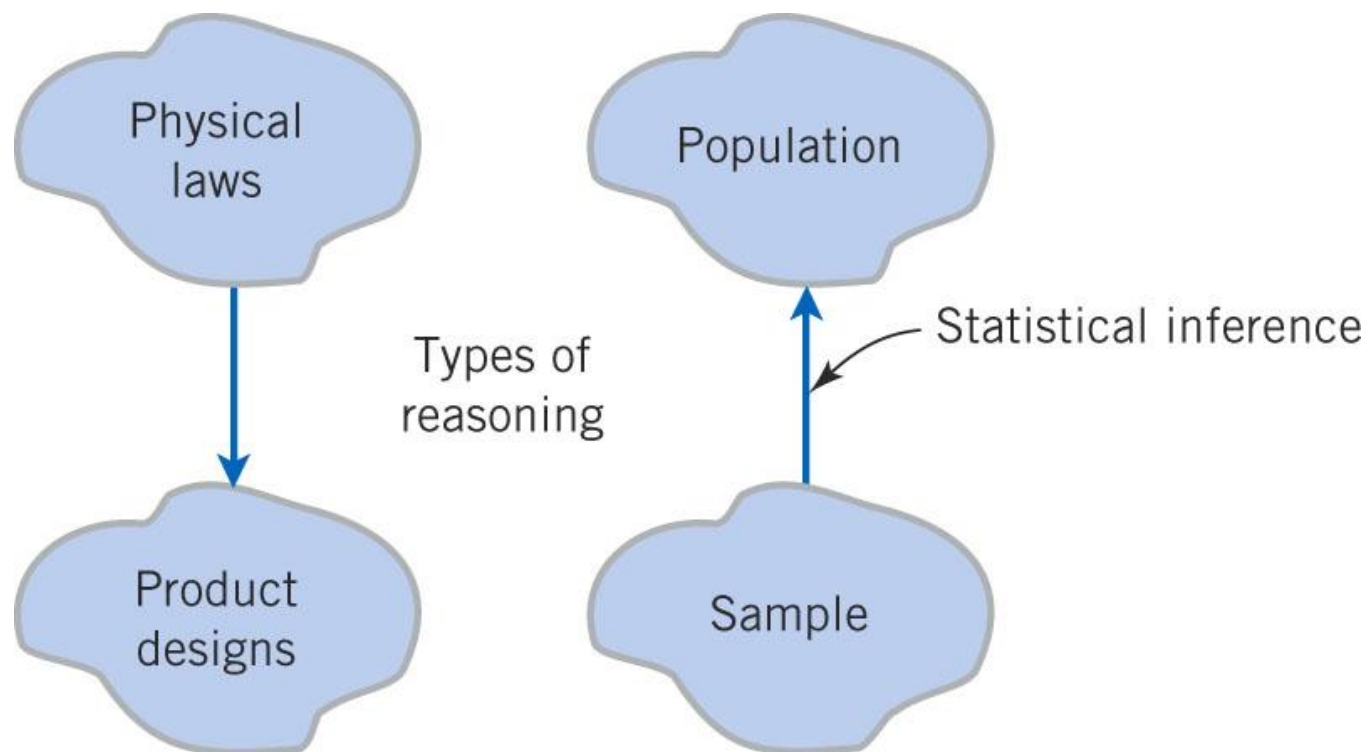
where μ is a constant and ε is a random disturbance.



Statistical Inference



Two Directions of Reasoning



Statistical inference is
one type of reasoning.

Basic Types of Studies

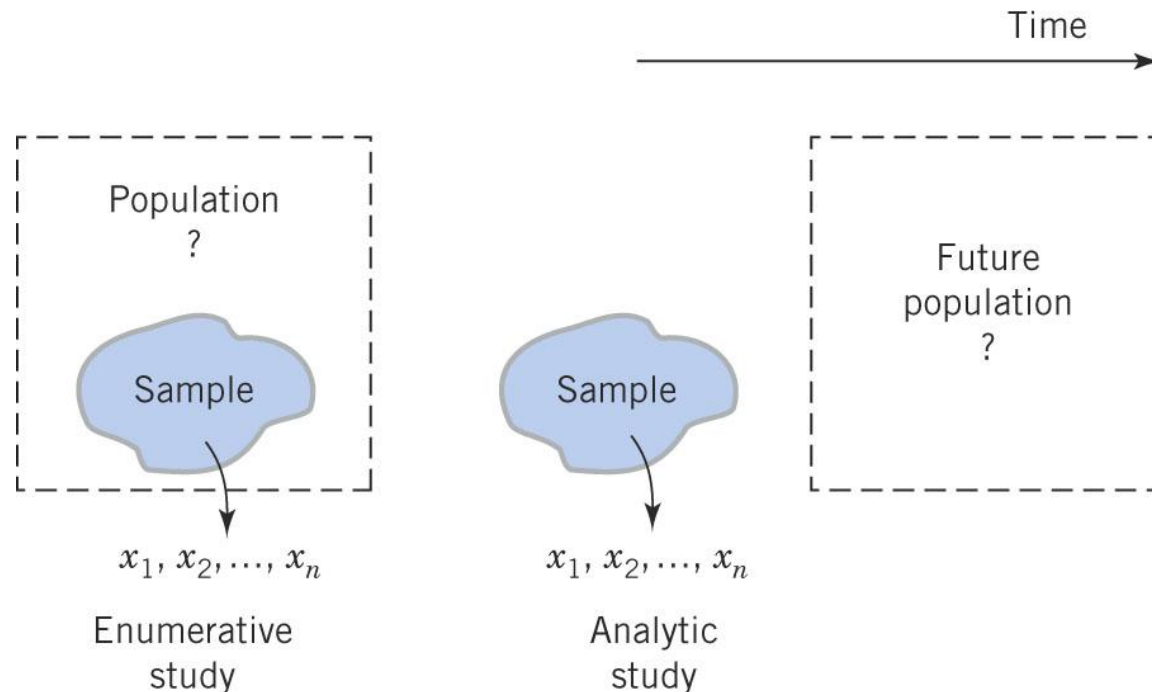
Three basic methods for collecting data:

- ♣ A **retrospective** study using historical data
 - ◆ Data collected in the past for other purposes.
- ♣ An **observational** study
 - ◆ Data presently collected by a passive observer.
- ♣ A **designed experiment**
 - ◆ Data collected in response to process input changes.



Random Samples

- ♣ Almost all statistic analysis is based on the using a **sample** of data that has been selected from some **population**.





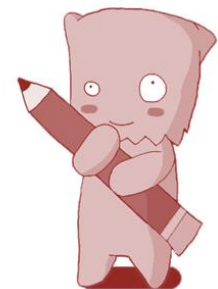
Hypothesis Tests

- ♣ A statement about a process behaviour value.
- ♣ Compared to a claim about another process value.
- ♣ Data is gathered to support or refute the claim.

- ♣ One-sample hypothesis test:
 - ◆ Ex. Ford avg mpg = 30 vs. avg mpg < 30
- ♣ Two-sample hypothesis test:
 - ◆ Ex. Ford avg mpg – Chevy avg mpg = 0 vs. > 0.



Time Series Data

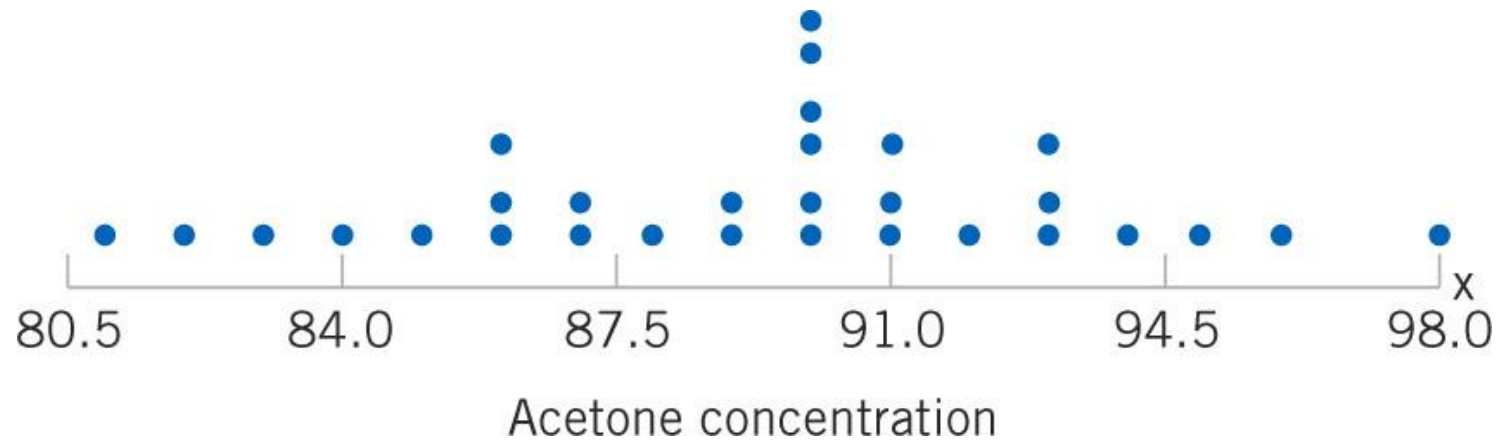




Observation Processes over Time

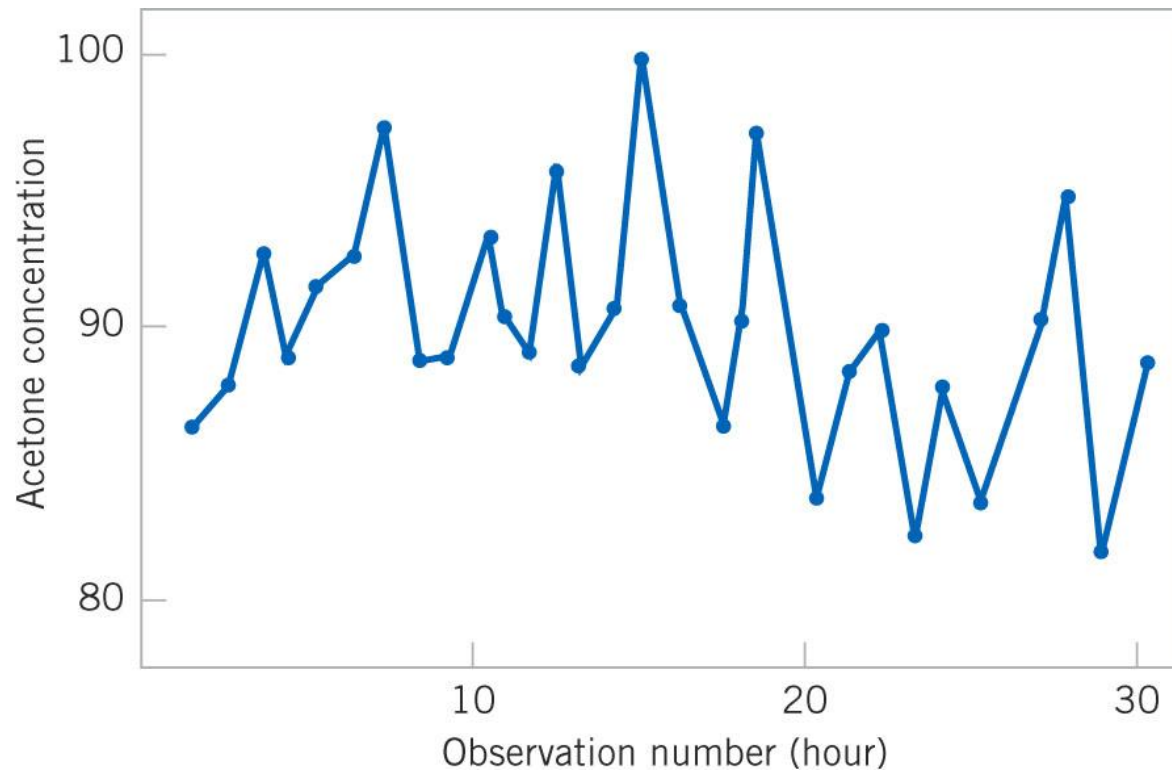
- ♣ Whenever data are collected **over time**, it is important to plot the data over time.
- ♣ Phenomena that might affect the system or process often become **more visible in a time-oriented plot** and the concept of stability can be better judged.

Observation Processes over Time (Cont.)



The dot diagram illustrates data centrality and variation, but does not identify any time-oriented problem.

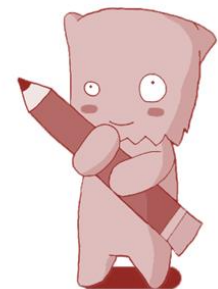
Observation Processes over Time (Cont.)



A time series plot of concentration provides more information than a dot diagram – shows a developing trend.



Prediction Model





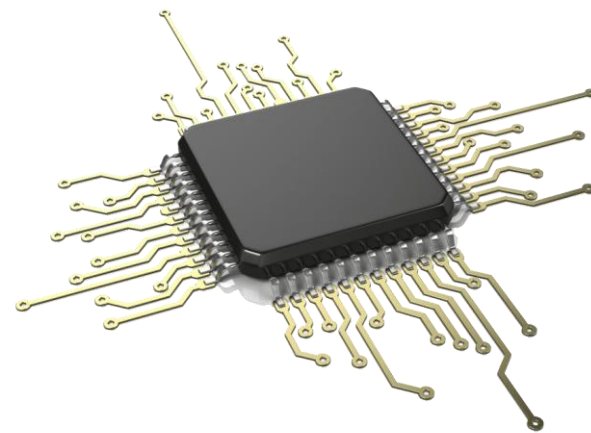
Mechanistic and Empirical Models

- ♣ An **empirical model** is built from our engineering and scientific knowledge of the phenomenon, but is not directly developed from our theoretical or first-principles understanding of the underlying mechanism.
- ♣ The form of the function is not known *a priori*.

Another Example of an Empirical Model

♣ In a semiconductor manufacturing plant, the finished semiconductor is wire-bonded to a frame. In an observational study, the variables recorded were:

- ◆ Pull strength to break the bond (y)
- ◆ Wire length (x_1)
- ◆ Die height (x_2)

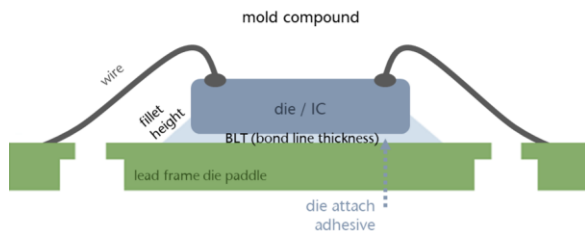


♣ The data recorded are shown on the next slide.



Wire Bond Pull Strength Data

Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2
1	9.95	2	50
2	24.45	8	110
3	31.75	11	120
4	35.00	10	550
5	25.02	8	295
6	16.86	4	200
7	14.38	2	375
8	9.60	2	52
9	24.35	9	100
10	27.50	8	300
11	17.08	4	412
12	37.00	11	400
13	41.95	12	500
14	11.66	2	360
15	21.65	4	205
16	17.89	4	400
17	69.00	20	600
18	10.30	1	585
19	34.93	10	540
20	46.59	15	250
21	44.88	15	290
22	54.12	16	510
23	56.63	17	590
24	22.13	6	100
25	21.15	5	400





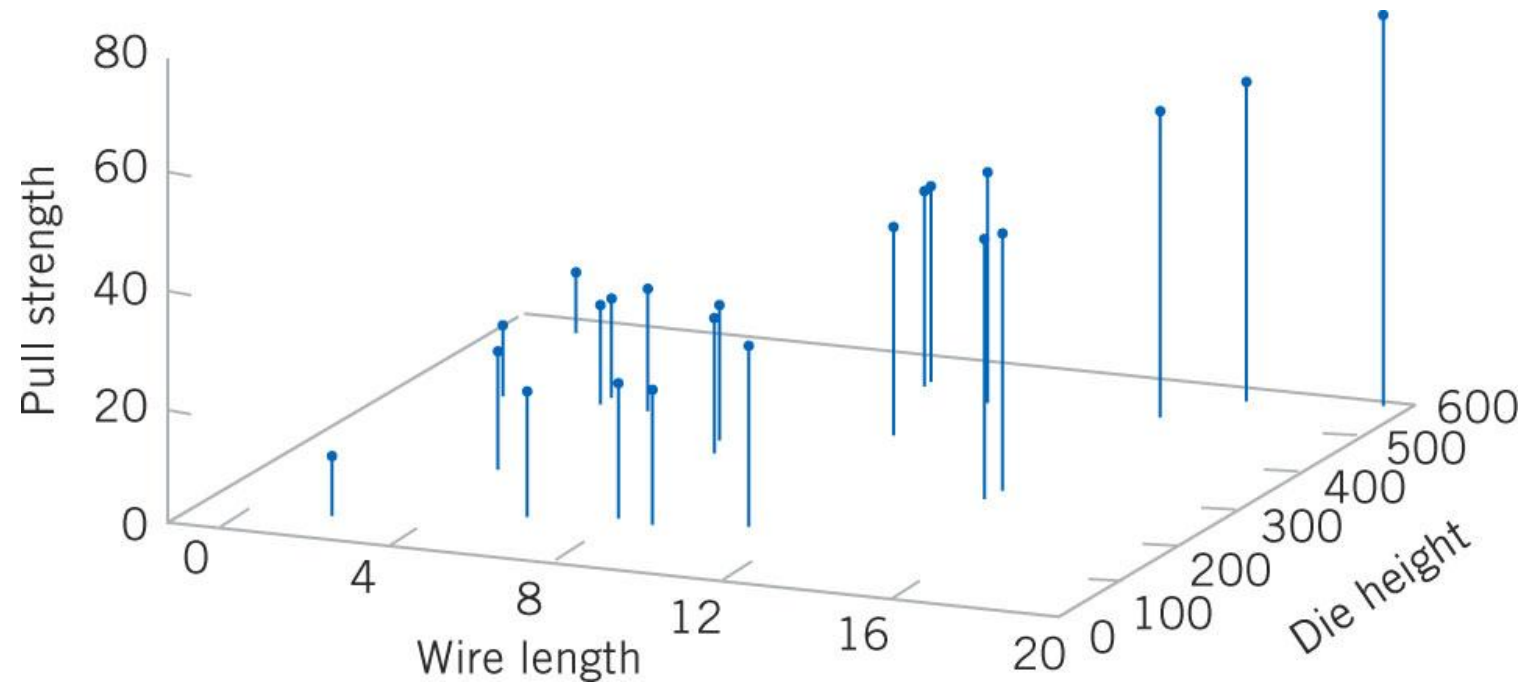
Empirical Model That Was Developed

- ♣ In general, this type of empirical model is called a **regression model**.

$$\text{Pull strength} = \beta_0 + \beta_1(\text{wire length}) + \beta_2(\text{die height}) + \varepsilon$$



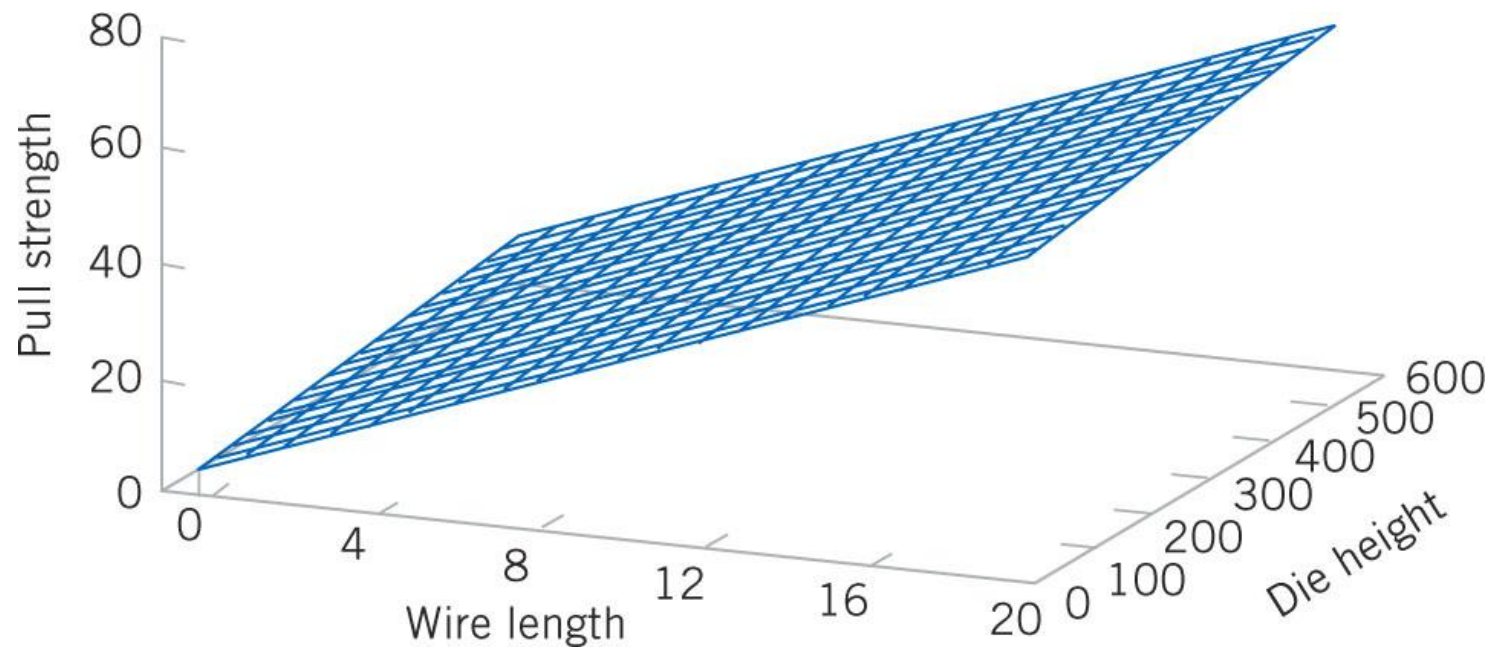
Visualizing the Data



Three-dimensional plot of the pull strength (y), wire length (x_1) and die height (x_2) data.



Visualizing the Resultant Model Using Regression Analysis



Plot of the predicted values (a plane) of pull strength from the empirical regression model.

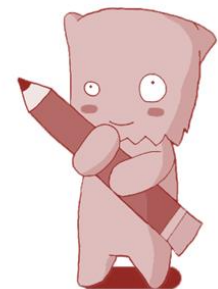


Models Can Also Reflect Uncertainty

- ♣ **Probability models** help quantify the risks involved in statistical inference, that is, risks involved in decisions made every day.
- ♣ Probability provides the **framework** for the study and application of statistics.



Factor Experiment



Factor Experiment Example

Consider a petroleum distillation column:

- ♣ Output is acetone concentration
- ♣ Inputs (factors) are:
 1. Reboil temperature
 2. Condensate temperature
 3. Reflux rate
- ♣ Output changes as the inputs are changed by experimenter.





Factor Experiment Example (Cont.)

- ♣ Each factor is set at **2 reasonable levels**
(-1 and +1)
- ♣ 8 (2^3) runs are made, at every combination of factors, to observe acetone output.
- ♣ Resultant data is used to create a mathematical model of the process representing cause and effect.

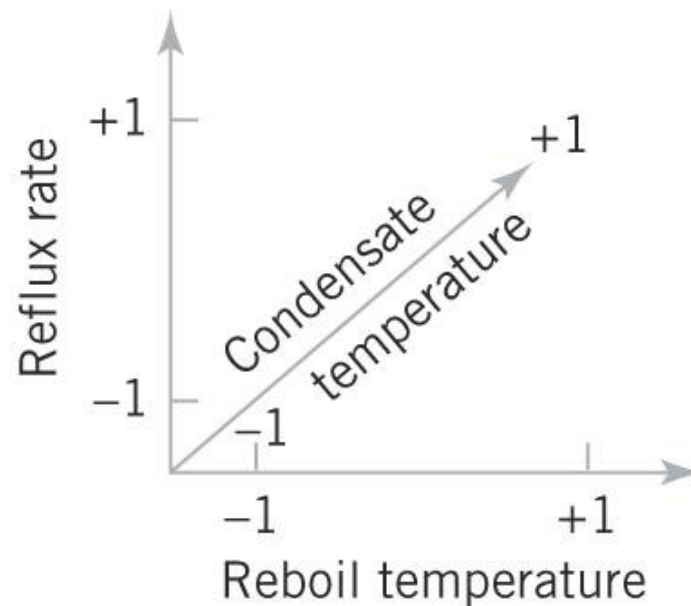
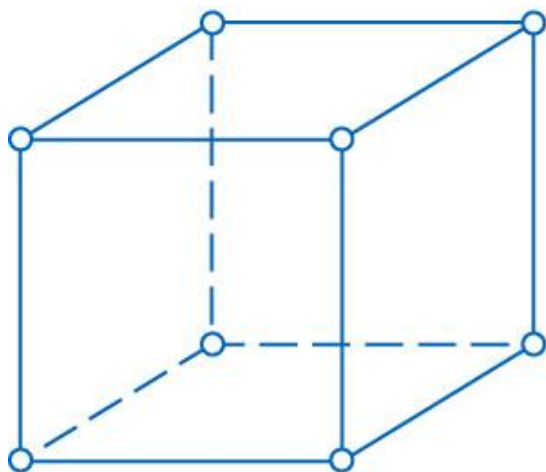


Factor Experiment Example (Cont.)

Reboil Temp.	Condensate Temp.	Reflux Rate
-1	-1	-1
+1	-1	-1
-1	+1	-1
+1	+1	-1
-1	-1	+1
+1	-1	+1
-1	+1	+1
+1	+1	+1

The Designed Experiment
(Factorial Design) for the
Distillation Column

Factor Experiment Example (Cont.)



The factorial experiment for the distillation column.

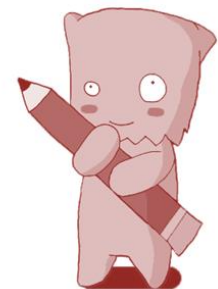
Factor Experiment Considerations

- ♣ Factor experiments can get too large. For example, 8 factors will require $2^8 = 256$ experimental runs of the distillation column.
- ♣ Certain combinations of factor levels can be **deleted** from the experiments without degrading the resultant model.
- ♣ The result is called a **fractional factorial experiment**.





Process Control



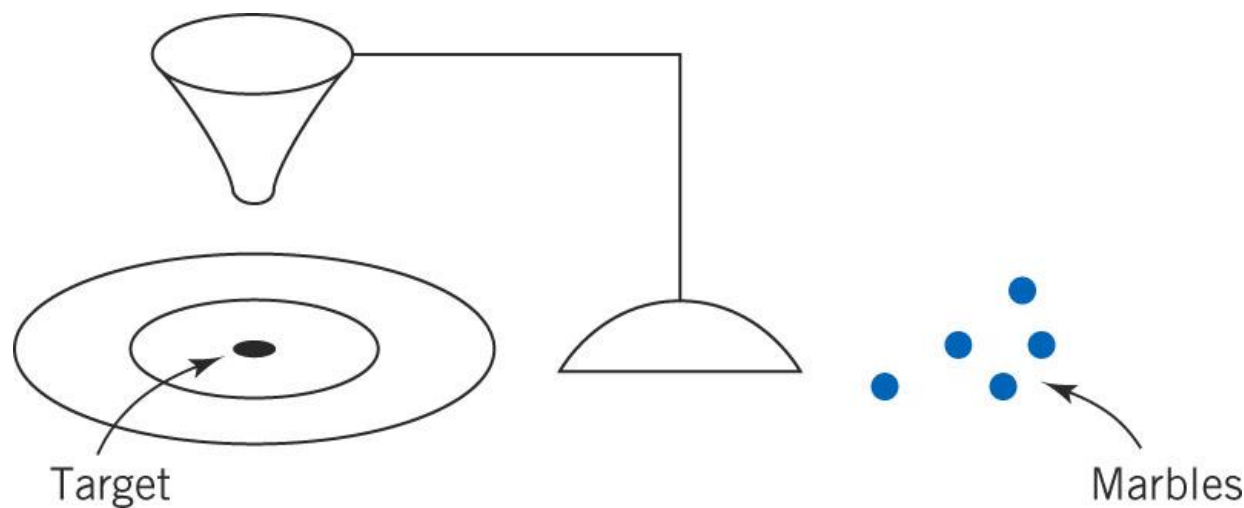


An Experiment in Variation

- ♣ W. Edwards Deming, a famous industrial statistician & contributor to the Japanese quality revolution, conducted a illustrative experiment on process **over-control** or **tampering**.
- ♣ Let's look at his apparatus and experimental procedure.

Deming's Experimental Set-up

- ♣ **Marbles** were dropped through a funnel onto a target and the location where the marble struck the target was recorded.



Deming's Funnel experiment



Deming's Experimental Set-up (Cont.)

- ♣ Variation was caused by several factors: marble placement in funnel & **release dynamics, vibration, air currents, measurement errors.**
- ♣ The funnel was aligned with the center of the target.
- ♣ Marbles were dropped.
- ♣ The distance from the strike point to the target center was measured and recorded.



Deming's Experimental Procedure

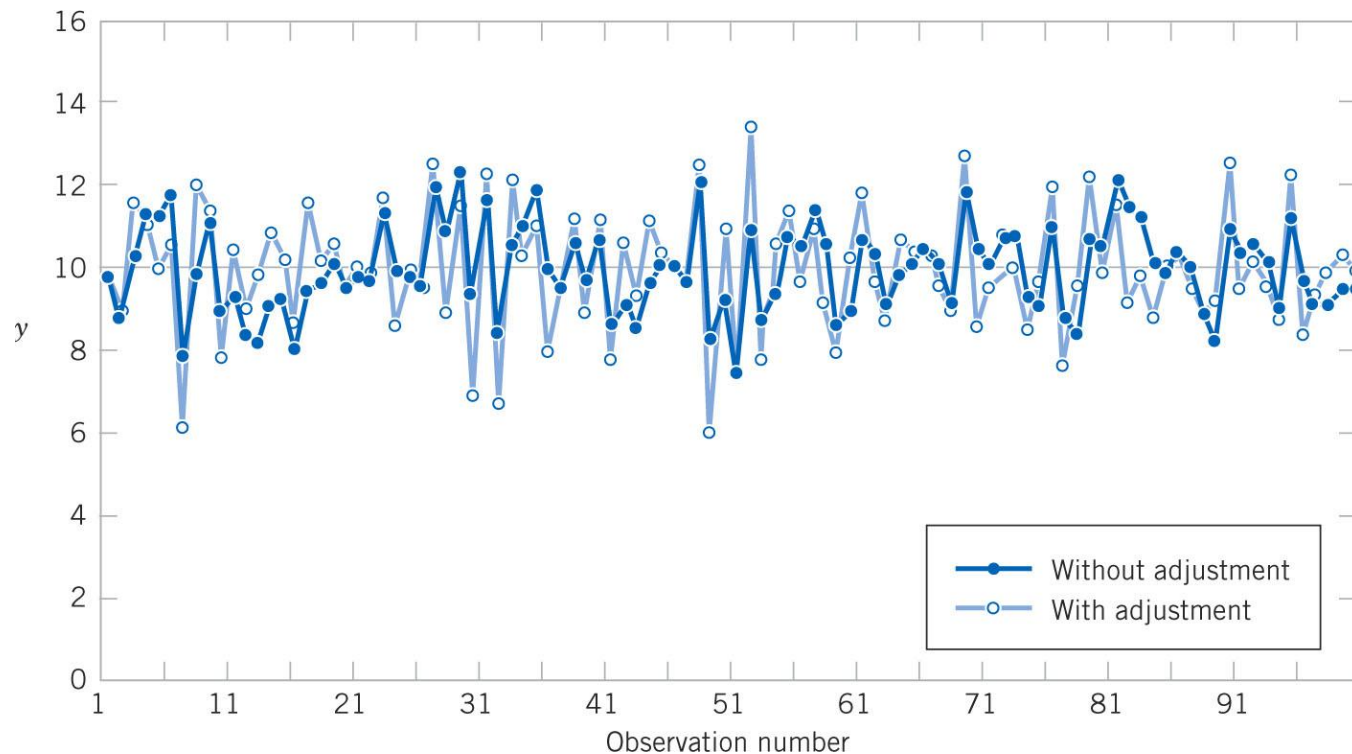
♣ Strategy 1:

- ◆ The funnel was not moved.
- ◆ Then the process was repeated.

♣ Strategy 2:

- ◆ The funnel was moved an equal distance in the opposite direction to compensate for the error.
- ◆ Then the process was repeated.

Adjustments Increased Variability



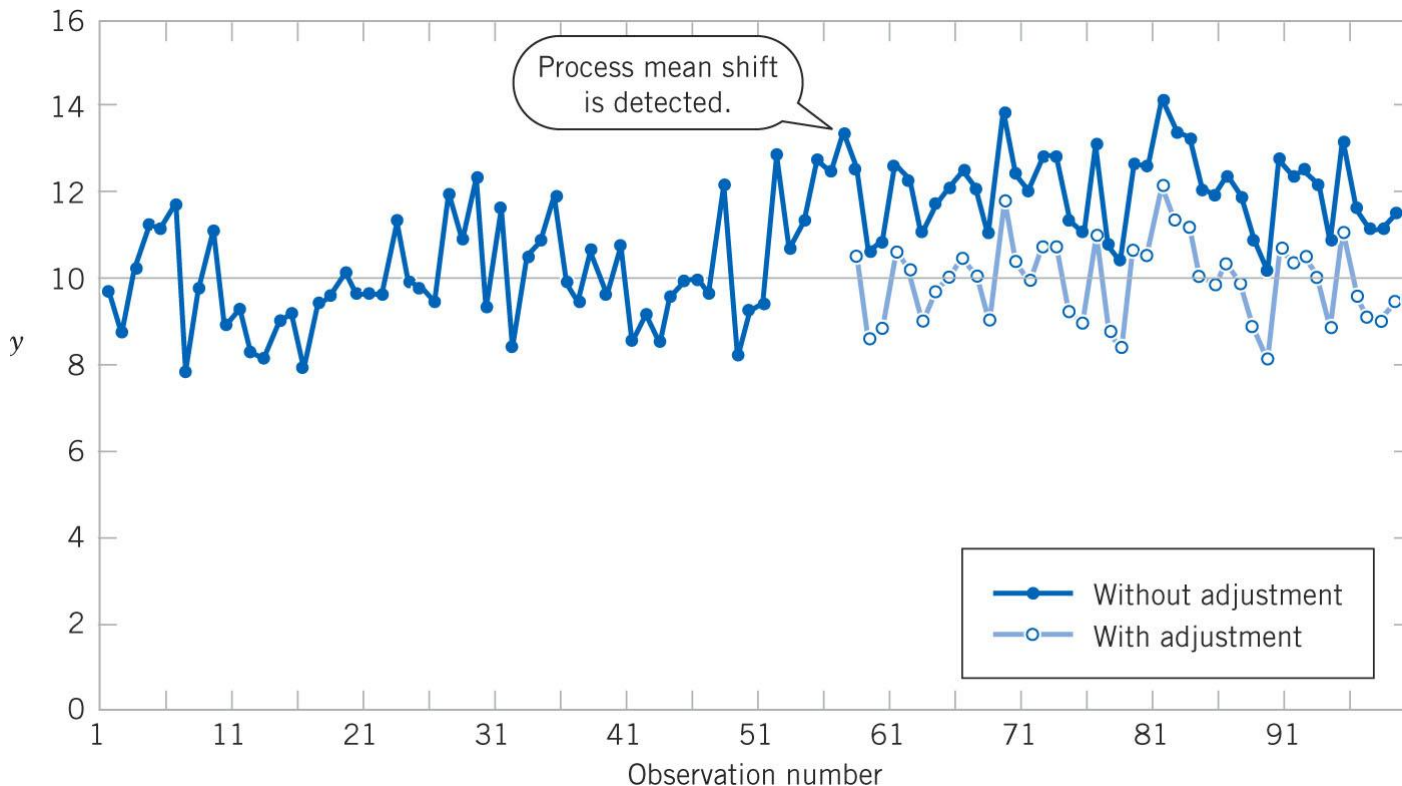
Adjustments applied to random disturbances over-controlled the process and increased the deviations from the target.



Conclusions from the Deming Experiment

- ♣ The lesson of the Deming experiment is that a process should not be adjusted in response to random variation, but only when a clear shift in the process value becomes apparent.
- ♣ To identify when the shift occurs, a control chart is used. Output values, plotted over time along with the outer limits of normal variation, pinpoint when the process leaves normal values and should be adjusted.

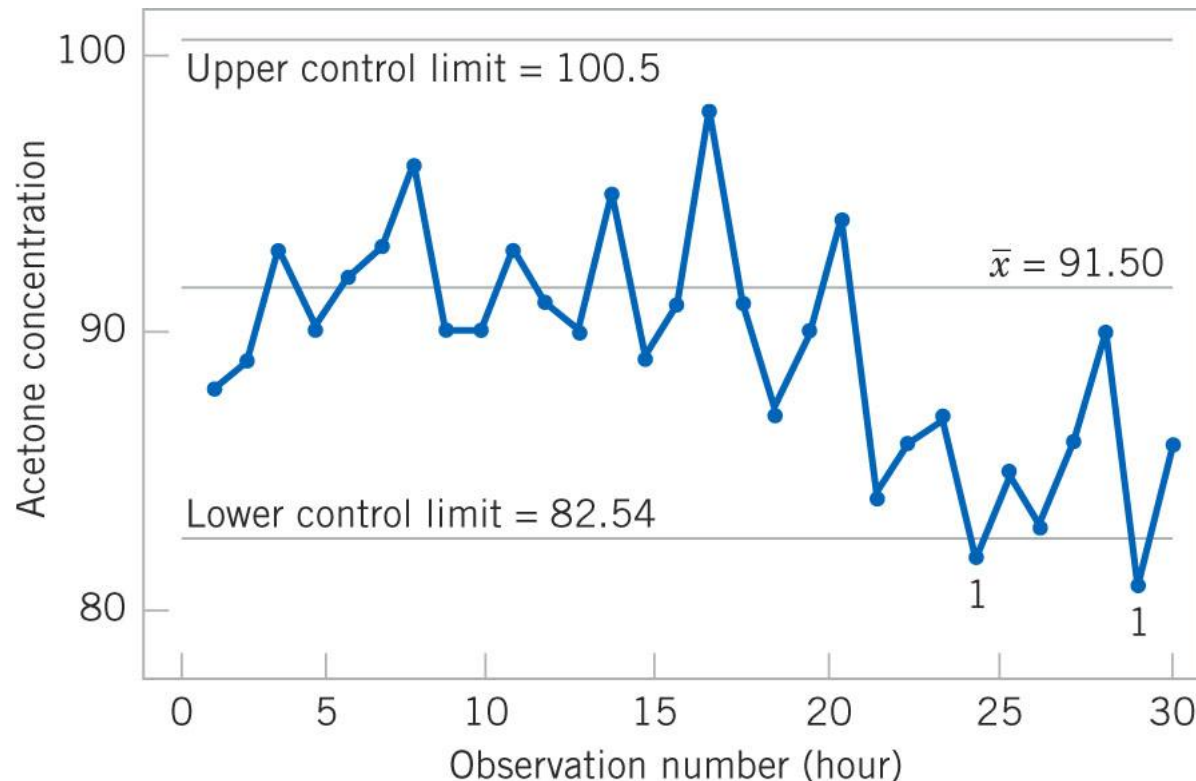
Detecting & Correcting the Process



Process mean shift is detected at observation #57, and an adjustment (a decrease of two units) reduces the deviations from target.



How Is the Change Detected Graphically?



A control chart for the chemical process concentration data.
Process steps out at hour 24 & 29. Shut down & adjust process.

